IO_VOUT.h File Reference

IO Driver functions for voltage outputs. More...

```
#include "IO_Driver.h"
```

Functions

IO_ErrorType	**IO_VOUT_Init** (**ubyte1** vout_channel) Setup one voltage output channel. More...
IO_ErrorType	**IO_VOUT_DeInit** (**ubyte1** vout_channel) Deinitializes one voltage output. More...
IO_ErrorType	**IO_VOUT_SetVoltage** (**ubyte1** vout_channel, **ubyte2** output_voltage) Sets the output voltage of one voltage output channel. More...
IO_ErrorType	**IO_VOUT_GetVoltage** (**ubyte1** vout_channel, **ubyte2** *const voltage, **bool** *const fresh) Get the voltage feedback of one voltage output channel. More...
IO_ErrorType	**IO_VOUT_ResetProtection** (**ubyte1** vout_channel, **ubyte1** *const reset_cnt) Reset the output protection for a VOUT channel. More...

Detailed Description

IO Driver functions for voltage outputs.

Contains all service functions for the voltage outputs. Up to 8 channels can be configured: `IO_VOUT_00` .. `IO_VOUT_07`

The voltage output stage is a push/pull PWM output with a well defined output resistance of 2.58kOhm. This resistance is necessary for operation, low pass filtering and over load protection.

Note

The voltage output is limited to resistive loads to ground with 10kOhm or higher.

The outputs will be activated after setting them via `IO_VOUT_SetVoltage()`.

When configuring a voltage output, the associated voltage feedback channel will also be configured.

Note

Deviations on the power supply are automatically corrected by the module.

VOUT-API Usage:

- **Examples for VOUT API functions**

VOUT output protection

Each voltage output is individually protected against malfunction. Whenever the difference between the measured output (U_feedback) and the configured output calculated by **U_diff = output_value / 100 * U_BAT - U_feedback** is greater than **abs(+/-18V)** for at least 100ms, the output protection is enabled latest within 12ms.

When entering the protection state, the voltage output has to remain in this state for at least 1s. After this wait time the voltage output can be reenabled via function `IO_VOUT_ResetProtection()`. Note that the number of reenabling operations for a single voltage output is limited to 10. Refer to function `IO_VOUT_ResetProtection()` for more information on how to reenable a voltage output.

VOUT Code Examples

Examples for using the VOUT API

VOUT initialization example:

```
IO_VOUT_Init(IO_VOUT_00);        // setup a voltage output
```

VOUT task function example:

```
ubyte2 voltage = 0;
bool fresh = FALSE;

IO_VOUT_SetVoltage(IO_VOUT_00,
                   6200);        // set output voltage to 6200mV

IO_VOUT_GetVoltage(IO_VOUT_00,
                   &voltage,
                   &fresh);      // get voltage feedback
```

Function Documentation

IO_ErrorType IO_VOUT_DeInit (ubyte1 vout_channel)

Deinitializes one voltage output.

Parameters

vout_channel VOUT output (`IO_VOUT_00` .. `IO_VOUT_07`)

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a VOUT channel
IO_E_UNKNOWN	an unknown error occurred

IO_ErrorType IO_VOUT_GetVoltage (ubyte1 vout_channel, ubyte2 *const voltage, bool *const fresh)

Get the voltage feedback of one voltage output channel.

Parameters

	vout_channel	VOUT channel (`IO_VOUT_00` .. `IO_VOUT_07`)
[out]	**voltage**	Measured voltage in mV. Range: 0..32000 (0V..32.000V)
[out]	**fresh**	Indicates if new values are available since the last call. • `TRUE`: Value in "voltage" is valid • `FALSE`: No new value available.

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_NULL_POINTER	a NULL pointer has been passed to the function
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a VOUT channel
IO_E_STARTUP	the output is in the startup phase
IO_E_REFERENCE	the reference voltage is out of range
IO_E_UNKNOWN	an unknown error occurred

Remarks

If there is no new measurement value available (for example the function `IO_VOUT_GetVoltage()` gets called more frequently than the AD sampling) the flag `fresh` will be set to `FALSE`.

IO_ErrorType IO_VOUT_Init (ubyte1 vout_channel)

Setup one voltage output channel.

Parameters

vout_channel VOUT channel (`IO_VOUT_00` .. `IO_VOUT_07`)

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist or is not available on the used ECU variant
IO_E_CH_CAPABILITY	the given channel is not a VOUT channel or the used ECU variant does not support this function
IO_E_CHANNEL_BUSY	the VOUT output channel or the voltage feedback channel is currently used by another function
IO_E_DRIVER_NOT_INITIALIZED	the common driver init function has not been called before
IO_E_FPGA_NOT_INITIALIZED	the FPGA has not been initialized
IO_E_UNKNOWN	an unknown error occurred

IO_ErrorType IO_VOUT_ResetProtection (ubyte1 vout_channel,
ubyte1 *const reset_cnt
)

Reset the output protection for a VOUT channel.

Parameters

	vout_channel	VOUT channel (`IO_VOUT_00` .. `IO_VOUT_07`)
[out]	**reset_cnt**	Protection reset counter. Indicates how often the application already reset the protection.

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_FET_PROT_NOT_ACTIVE	no output FET protection is active
IO_E_FET_PROT_PERMANENT	the output FET is protected permanently because it was already reset more than 10 times
IO_E_FET_PROT_WAIT	the output FET protection can not be reset, as the wait time of 1s is not already passed
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a VOUT channel
IO_E_UNKNOWN	an unknown error occurred

Attention

The protection can be reset 10 times, afterwards the output will remain permanently protected

Note

- After entering the output protection, a certain time has to pass before the output protection can be reset:
  - 1s for `IO_VOUT_00` .. `IO_VOUT_07`
- This function will not set the output back again to the last voltage. After calling this function the output has to be set to the intended voltage with `IO_VOUT_SetVoltage()`

Remarks

If the parameter `reset_cnt` is `NULL`, the parameter is ignored. The parameter `reset_cnt` returns the number of resets already performed.

**IO_ErrorType IO_VOUT_SetVoltage (ubyte1 vout_channel,
ubyte2 output_voltage
)**

Sets the output voltage of one voltage output channel.

Parameters

vout_channel	VOUT channel (`IO_VOUT_00` .. `IO_VOUT_07`)
output_voltage	Output voltage in mV (0..32000)

Returns

IO_ErrorType

Return values

IO_E_OK	everything fine
IO_E_INVALID_CHANNEL_ID	the given channel id does not exist
IO_E_FET_PROT_ACTIVE	the output FET protection is active and has set the output to 0V. It can be reset with `IO_VOUT_ResetProtection()` after the wait time is passed
IO_E_FET_PROT_REENABLE	the output FET protection is ready to be reset with `IO_VOUT_ResetProtection()`
IO_E_FET_PROT_PERMANENT	the output FET is protected permanently because it was already reset more than 10 times. The output will remain at 0V
IO_E_INVALID_PARAMETER	the setpoint is out of range
IO_E_CHANNEL_NOT_CONFIGURED	the given channel is not configured
IO_E_CH_CAPABILITY	the given channel is not a VOUT channel
IO_E_UNKNOWN	an unknown error occurred